



E/I/I test tables

Rejected take-off and runway overrun, Tarlton Springs

Exported 5 October 2026

INVESTIGATION WORKFLOW SUITE

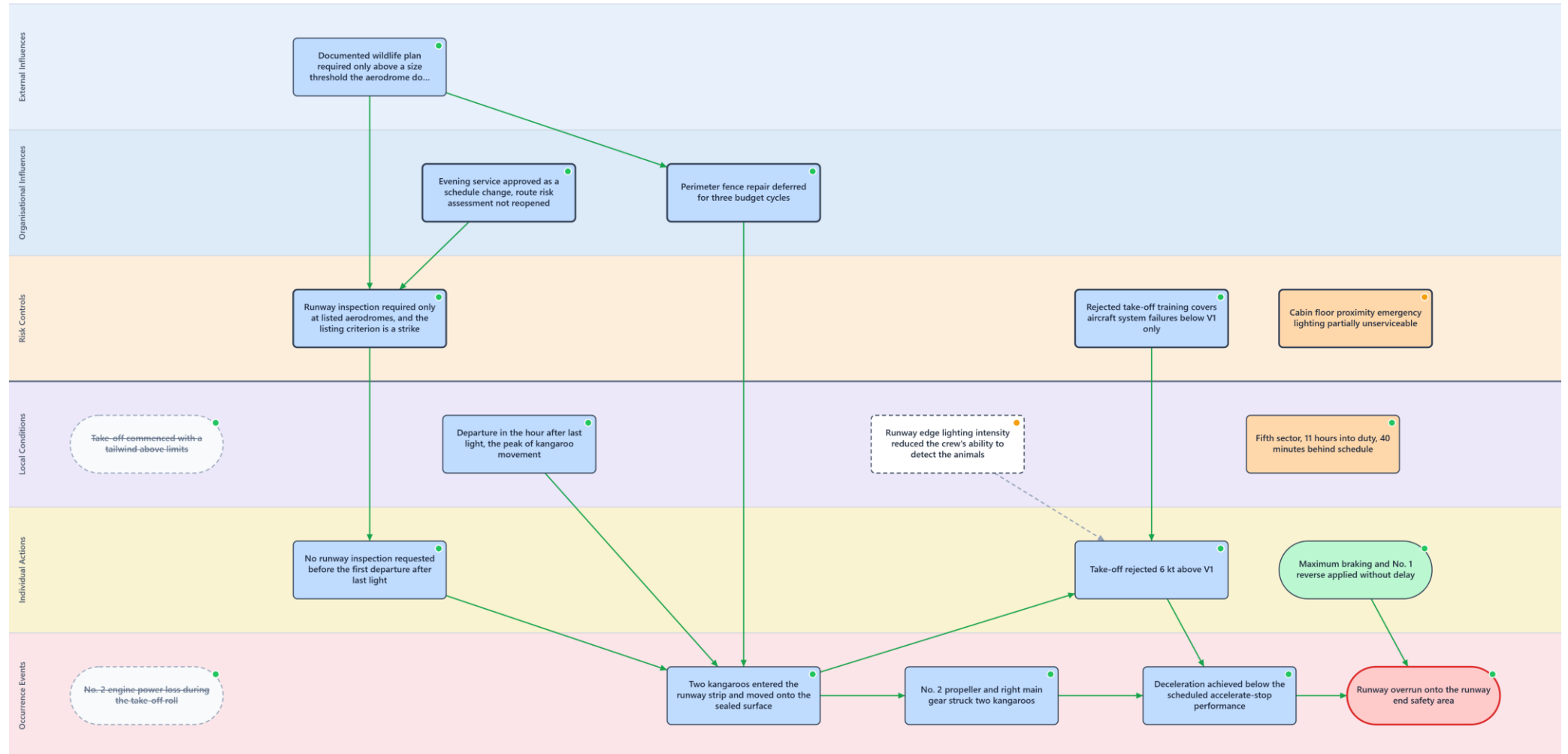
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AcciMap

Rejected take-off and runway overrun, Tarlton Springs

Rendered 03/10/2026, 12:06:31 pm



Existence: did it exist or occur? Influence: did it influence this occurrence? Importance: is it worth addressing regardless?

External Influences

Documented wildlife plan required only above a size threshold the aerodrome does not meet

Test	Result	Supporting text
Existence	Pass - virtually certain (100%)	AS-139 10.4.2 requires a documented wildlife hazard management plan only where the aerodrome is used by aeroplanes above 27,000 kg MTOW or serves more than 350,000 passenger movements a year. Tarlton Springs' heaviest scheduled aeroplane is 22,900 kg and it handled 61,400 passenger movements in 2025, so neither limb applies. Under 10.4.3 its measures need not be documented at all. The aerodrome accordingly held no plan, no hazard assessment, no dispersal programme and no record of measures beyond the single pre-first-arrival inspection at Aerodrome Manual 6.3. Confirmed against the certified aerodrome register and the aerodrome operator's own compliance file; the aerodrome is not in breach.
Influence	Pass - likely (70%)	The absence of a documented plan is the absence of the one mechanism that would have connected the aerodrome's own sighting log to the operators using the aerodrome, and the one that would have put the failed fence on a funded programme rather than an unfunded works register.
Importance	Not tested	(none)

Investigator notes: The aerodrome is not in breach of anything. That is what makes this an External Influences object rather than a compliance finding: the standard's own threshold is what leaves an aerodrome of this size with an undocumented, unaudited and in practice unfunded wildlife programme.

Evidence attached

10.4.1 The operator of a certified aerodrome must take reasonable measures to reduce the risk to aircraft operations presented by wildlife on or in the vicinity of the aerodrome.

10.4.2 The operator of a certified aerodrome must prepare, implement and maintain a documented wildlife hazard management plan where the aerodrome is used by aeroplanes having a maximum take-off weight exceeding 27,000 kg, or serves more than 350,000 passenger movements in a calendar year.

10.4.3 Where subsection 10.4.2 does not apply, the measures taken under 10.4.1 need not be documented, but the aerodrome operator must be able to demonstrate what measures are in place.

Tarlton Springs is a certified aerodrome. Its heaviest scheduled aeroplane is the 22,900 kg twin-turboprop used on the Saltbush Air service, and it handled 61,400 passenger movements in 2025. Section 10.4.2 therefore does not apply to it, and it held no wildlife hazard management plan. It is not in breach of the standard.

Source: Aerodrome Standard AS-139, section 10.4

46 entries. 41 of the 46

record kangaroos on or adjacent to the movement area. 38 of those 41 are

timed between one hour before and two hours after last light. Eleven entries record animals on the sealed surface of runway 09/27. The log is a paper register held in the aerodrome office. It is not reported to any operator, and the aerodrome reporting officer confirmed that no airline has ever asked for it.

Source: Tarlton Springs Aerodrome wildlife sighting log, copied 23/06/2026

Item WR-221, raised 04/2023: "Perimeter fence, northern boundary between the 700 m and 1,300 m runway markers, four panels displaced and wire strained, animals observed passing through. Recommend replacement, estimated \$41,000." Deferred at the 2023, 2024 and 2025 budget rounds, each time with the entry "defer, unfunded, review next cycle". Item WR-263, raised 02/2026, records the same section as "deteriorating further". A site inspection on 23/06/2026 found a continuous 140 m section of the northern fence with the bottom two wires separated from the posts and a clear animal pad through it at the 1,050 m mark.

Source: Tarlton Shire Council aerodrome works register WR-221 (04/2023), WR-263 (02/2026)

Organisational Influences

Evening service approved as a schedule change, route risk assessment not reopened

Test	Result	Supporting text
Existence	Pass - virtually certain (100%)	RRA-041 issue 3 (11/2022) is the current route risk assessment for Tarlton Springs. It carries the operating restriction 'daylight operations only pending review of night lighting and diversion fuel policy'. CMR-2025-118 (approved by the Head of Flying Operations on 22/05/2025) introduced a scheduled evening rotation from 14/07/2025, the first company operation into Tarlton Springs after last light. It assessed the change as 'schedule change only', recorded 'no new hazard identified', and answered 'No' to route risk assessment review required. RRA-041 was not reopened and no issue 4 exists. The Safety Department's records show no involvement beyond receipt of the approved record.
Influence	Pass - likely (85%)	The change put a company aircraft into an aerodrome after last light for the first time, in explicit contradiction of a live restriction, and did so through a process that asked whether the change was new rather than whether the assessment still held.
Importance	Not tested	(none)

Investigator notes: The clearest single organisational finding in the investigation. The restriction that should have blocked this change was sitting in the operator's own current route risk assessment, and the change record does not mention it.

Evidence attached

8.14.1 Wildlife presents a hazard to aircraft operations, particularly at regional aerodromes and particularly around dawn and dusk.

8.14.2 At an aerodrome listed in Appendix 8A, the pilot in command shall request a runway inspection from the aerodrome operator prior to the first departure after last light, and shall not commence the take-off roll until the result of that inspection has been passed.

8.14.3 At all other aerodromes, no specific action is required. Crews should remain alert to the possibility of wildlife on the movement area.

8.14.4 A wildlife strike shall be reported on an air safety report. Where a strike occurs, the Safety Department will review the aerodrome's status against the criteria in Appendix 8A.

Appendix 8A, Wildlife-active aerodromes. An aerodrome is listed in this appendix where a wildlife strike involving a company aircraft has been reported at that aerodrome within the preceding 24 months. The list is reviewed annually by the Safety Department.

Appendix 8A as current on 18/06/2026 lists nine aerodromes. Tarlton Springs is not among them, and has never been listed. The list was last reviewed on 03/03/2025.

The gap. The only route onto the list is a reported strike. A sighting

cannot put an aerodrome on the list, and nothing in Part A asks a crew to report a sighting. Tarlton Springs has had no company strike, so it cannot be listed; because it is not listed, no inspection is requested; because no inspection is requested, nothing is looked for. The control is triggered by the outcome it exists to prevent.

Source: Saltbush Air Operations Manual Part A section 8.14, revision 9, 04/2024

Hazards assessed: runway length and accelerate-stop margin; single runway with no lighting redundancy; no alternate within 200 nm at night; limited engineering support; heat and density altitude in summer.

Operating restriction: daylight operations only pending review of night lighting and diversion fuel policy.

Issue 3 is the current issue. No issue 4 exists.

6. Change Management Record CMR-2025-118, "Introduction of the Tarlton Springs evening service", 05/2025

Scope of change: addition of one evening rotation, Sydney to Tarlton Springs to Sydney, six days per week, from 14/07/2025.

Assessment: schedule change only. Existing aerodrome, existing aircraft type, existing crew qualifications, existing engineering support arrangements. No new hazard identified.

Route risk assessment review required: No.

Approved: Head of Flying Operations, 22/05/2025.

The change introduced the first scheduled operation into Tarlton Springs after last light. The daylight-only restriction at RRA-041 issue 3 is not referenced anywhere in CMR-2025-118, and RRA-041 was not reopened. The Safety Department's records show no involvement in the change beyond receipt of the approved record.

Source: RRA-041 issue 3 (11/2022) and CMR-2025-118 (approved 22/05/2025)

Perimeter fence repair deferred for three budget cycles

Test	Result	Supporting text
Existence	Pass - virtually certain (100%)	Works register item WR-221, raised 04/2023: four displaced panels and strained wire on the northern boundary between the 700 m and 1,300 m runway markers, with animals observed passing through, replacement estimated at \$41,000. Deferred at the 2023, 2024 and 2025 budget rounds, each entry reading 'defer, unfunded, review next cycle'. WR-263 (02/2026) records the same section deteriorating further. The investigator's site inspection on 23/06/2026 found a continuous 140 m section with the bottom two wires separated from the posts and a clear animal pad through the fence at the 1,050 m mark, 110 m before the point at which the first carcass was recovered.
Influence	Pass - very likely (90%)	This is the physical route by which the animals reached the movement area. The pad through the fence at 1,050 m and the carcasses at 1,188 m and 1,221 m are 140 m apart on the same side of the runway.
Importance	Not tested	(none)

Investigator notes: The deferrals are recorded in the aerodrome operator's own register in its own words, including 'animals observed passing through' in the original 2023 entry. Nothing here had to be reconstructed.

Evidence attached

46 entries. 41 of the 46

record kangaroos on or adjacent to the movement area. 38 of those 41 are timed between one hour before and two hours after last light. Eleven entries record animals on the sealed surface of runway 09/27. The log is a paper register held in the aerodrome office. It is not reported to any operator, and the aerodrome reporting officer confirmed that no airline has ever asked for it.

Source: Tarlton Springs Aerodrome wildlife sighting log, copied 23/06/2026

Item WR-221, raised 04/2023: "Perimeter fence,

northern boundary between the 700 m and 1,300 m runway markers, four panels displaced and wire strained, animals observed passing through. Recommend replacement, estimated \$41,000." Deferred at the 2023, 2024 and 2025 budget rounds, each time with the entry "defer, unfunded, review next cycle". Item WR-263, raised 02/2026, records the same section as "deteriorating further".

A site inspection on 23/06/2026 found a continuous 140 m section of the northern fence with the bottom two wires separated from the posts and a clear animal pad through it at the 1,050 m mark.

Source: Tarlton Shire Council aerodrome works register WR-221 (04/2023), WR-263 (02/2026)



The northern perimeter fence at the 1,050 m mark, photographed on the site visit: the fallen section and the animal pad through it. Site inspection photograph SV-2026-0417-07, northern boundary at the 1,050 m runway marker, taken 23/06/2026

Risk Controls

Runway inspection required only at listed aerodromes, and the listing criterion is a strike

Test	Result	Supporting text
Existence	Pass - virtually certain (100%)	Operations Manual Part A 8.14.2 requires the pilot in command to request a runway inspection before the first departure after last light at any aerodrome listed in Appendix 8A, and not to commence the roll until the result is passed. 8.14.3 requires nothing at other aerodromes. Appendix 8A lists an aerodrome where a wildlife strike involving a company aircraft has been reported there in the preceding 24 months. Nine aerodromes are listed; Tarlton Springs never has been. No provision anywhere in Part A asks a crew to report a sighting, and no provision admits an aerodrome to the list on any other evidence. The list was last reviewed on 03/03/2025, four months before the evening service began.
Influence	Pass - very likely (90%)	This is the operator's own defence against exactly this occurrence, and it was unarmed. Eleven sightings of kangaroos on the sealed surface at this aerodrome could not put it on the list, because they were not strikes and were not reported to the company.
Importance	Not tested	(none)

Investigator notes: The safety issue the map exists to find. The control is sound and the trigger is circular: it can only be armed by the event it is there to prevent. Recommend this one in its own right regardless of anything else in the report.

Evidence attached

8.14.1 Wildlife presents a hazard to aircraft operations, particularly at regional aerodromes and particularly around dawn and dusk.

8.14.2 At an aerodrome listed in Appendix 8A, the pilot in command shall request a runway inspection from the aerodrome operator prior to the first departure after last light, and shall not commence the take-off roll until the result of that inspection has been passed.

8.14.3 At all other aerodromes, no specific action is required. Crews should remain alert to the possibility of wildlife on the movement area.

8.14.4 A wildlife strike shall be reported on an air safety report. Where a strike occurs, the Safety Department will review the aerodrome's status against the criteria in Appendix 8A.

Appendix 8A, Wildlife-active aerodromes. An aerodrome is listed in this appendix where a wildlife strike involving a company aircraft has been reported at that aerodrome within the preceding 24 months. The list is reviewed annually by the Safety Department.

Appendix 8A as current on 18/06/2026 lists nine aerodromes. Tarlton Springs is not among them, and has never been listed. The list was last reviewed on 03/03/2025.

The gap. The only route onto the list is a reported strike. A sighting

cannot put an aerodrome on the list, and nothing in Part A asks a crew to report a sighting. Tarlton Springs has had no company strike, so it cannot be listed; because it is not listed, no inspection is requested; because no inspection is requested, nothing is looked for. The control is triggered by the outcome it exists to prevent.

Source: Saltbush Air Operations Manual Part A section 8.14, revision 9, 04/2024
46 entries. 41 of the 46

record kangaroos on or adjacent to the movement area. 38 of those 41 are timed between one hour before and two hours after last light. Eleven entries record animals on the sealed surface of runway 09/27. The log is a paper register held in the aerodrome office. It is not reported to any operator, and the aerodrome reporting officer confirmed that no airline has ever asked for it.

Source: Tarlton Springs Aerodrome wildlife sighting log, copied 23/06/2026

Rejected take-off training covers aircraft system failures below V1 only

Test	Result	Supporting text
Existence	Pass - extremely likely (95%)	TRG-SYL-07 module 4's four scripted scenarios are engine failure at 90 kt, engine fire at 100 kt, configuration warning at 85 kt, and engine failure at V1 plus 5 kt continued. All four are aircraft system events and three are below V1. No crew has been presented with an external obstruction scenario in the five training cycles on file, covering 214 crew in the current cycle alone. Part B 4.6 matches: 4.6.3 lists reject items to V1, 4.6.4 states the take-off shall be continued at or above V1, and the section contains no guidance on an obstruction on the runway at any speed. Both the pilot in command and the first officer stated independently that every stop or go scenario they have been given has been an engine and has arrived below V1.
Influence	Pass - likely (75%)	The captain had two seconds and nothing behind him. The decision he made was the first time he had considered the question at all, which is what the training system is supposed to prevent.
Importance	Not tested	(none)

Investigator notes: Both crew raised this unprompted, in almost the same words, and the syllabus bears them out. Worth being careful in the report that this explains the decision rather than excusing it.

Evidence attached

4.6.2 Below 80 kt the take-off shall be rejected for any malfunction or condition that in the judgement of the pilot in command makes the aircraft unfit to fly.

4.6.3 Between 80 kt and V1 the take-off shall be rejected only for: fire or fire warning; engine failure; a configuration warning; loss of directional control; or the pilot in command's judgement that the aircraft is unsafe or unable to fly.

4.6.4 At or above V1 the take-off shall be continued. The decision to continue is made before V1 is reached.

Section 4.6 contains no guidance on an obstruction seen on the runway, at any speed, and no amplification of 4.6.4 for a case in which continuing is itself hazardous.

Source: Saltbush Air Operations Manual Part B section 4.6, revision 14, 09/2025

Module 4, Rejected take-off. Objective: the crew will demonstrate a correctly executed rejected take-off for an engine failure, an engine fire and a configuration warning, each initiated below V1, and will demonstrate the decision to continue for a failure recognised at or after V1.

The module's four scripted simulator scenarios are: engine failure at 90 kt; engine fire at 100 kt; configuration warning at 85 kt; engine failure at V1 plus 5 kt, continued. Every scenario is an aircraft system event. None presents an obstruction, an animal or any external hazard on the runway, at any speed.

The company's training records show 214 crew completed module 4 in the

2025 to 2026 cycle. No crew has been presented with an external obstruction scenario in the simulator in the five cycles held on file.

Source: TRG-SYL-07, cycle 2025 to 2026, module 4

Cabin floor proximity emergency lighting partially unserviceable

Test	Result	Supporting text
Existence	Pass - virtually certain (100%)	Raised by the first officer, who noticed sections of the aisle floor strip unlit while walking back through the cabin during the disembarkation. The functional check on 24/06/2026 found two of the six floor proximity escape path segments aft of row 16 on the left side dark on battery power with the system armed, traced to a failed common ballast. The pre-flight emergency lighting test on this type illuminates the ceiling lights and exit signs and neither illuminates nor monitors the floor segments, so the defect was undetectable by the crew or by a line engineer without a dedicated functional test. The last such test was 11 months before the occurrence, inside the 24-month programme interval.
Influence	Fail - unlikely (10%)	A partially dark escape path is a defence that is not there when it is needed, and nothing available to the crew would have told them.
Importance	Pass - very likely (90%)	It did not influence this occurrence and it is still a finding. The disembarkation was unhurried, by the forward airstair, with the main cabin lights on. Change one thing - a fire indication, smoke, a night evacuation over the wing - and two dark segments in the aft cabin are the difference between an orderly exit and a blocked one. The gap that matters is the monitoring; a 24-month interval on a system the daily test cannot see means the fleet's true state is unknown, and the same ballast is fitted across the type.

Investigator notes: Left in draft deliberately. The fleet-wide question is out with engineering and the object should not be finalised until that answer is in, even though its own tests are settled. No arrow, deliberately: Influence failed at 10%. It did not contribute to the overrun or to its outcome, because no evacuation was commanded and the main cabin lights were on throughout; it is carried on Importance alone, as a finding about a defence that was not there.

Evidence attached

Raised following the first officer's account. With the emergency lighting system armed and the aircraft on battery power, two of the six floor proximity escape path segments on the left side aft of row 16 did not illuminate. The fault was traced to a failed ballast common to both segments.

The pre-flight emergency lighting test on this type illuminates the ceiling emergency lights and the exit signs; it does not illuminate or monitor the floor proximity segments, so the defect was not detectable by any check available to the crew or to a line engineer without a dedicated functional test. The last dedicated functional test was 11 months before the occurrence, within the maintenance programme's 24-month interval.

The occurrence disembarkation was conducted with the main cabin lights on, by the forward airstair, without urgency. Nothing about the defect affected what happened on the night.

Source: Functional check record, VH-SBQ cabin emergency lighting, 24/06/2026

Local Conditions

Take-off commenced with a tailwind above limits

Test	Result	Supporting text
Existence	Fail - very unlikely (5%)	Hypothesis tested because a stop that runs 62 m past the end invites a question about the wind. It fails. The 06:30, 07:00 and 08:00 UTC observations give 090 at 7, 090 at 6 and 090 at 5 kt, a headwind component of 5 to 7 kt on runway 09. Recorded FDR wind during the roll gives a headwind component of 4 to 6 kt throughout, with no tailwind component at any point. The crew's performance entry used a 5 kt headwind, which matches. There is no evidence of a tailwind at any stage of the roll, from any source.
Influence	Not tested	(none)
Importance	Not tested	(none)

Investigator notes: Raised early because the overrun distance invited it. Disposed of in an afternoon and kept on the map so the report can say it was tested. No arrow, deliberately: Existence failed, so there is no event here to draw an arrow from. The wind did not contribute to the overrun because there was no tailwind at any point in the roll.

Evidence attached

UTC	Event	IAS	Groundspeed	Torque 1 / 2	Brake pressure	Notes
07:30:02	Power set, roll commences	0	0	98.4% / 98.5%	0 psi	Both power levers advanced together
07:30:15	80 kt cross-check	80	77	98.4% / 98.5%	0 psi	No caution or warning active
07:30:29	V1	112	109	98.4% / 98.5%	0 psi	Recorded aural "V1" on CVR
07:30:31	First officer's sighting call	116	113	98.4% / 98.5%	0 psi	No control input yet
07:30:32	Reject initiated	118	115	98.4% / 98.5%	0 psi	Both power levers to flight idle within 0.6 s
07:30:33	Impact	114	111	12% / 9%	2,180 psi	Lateral acceleration spike 0.31 g right; vertical spike 1.48 g
07:30:34	Maximum braking established	109	106	6% / 6%	2,940 psi	Anti-skid active on all four wheels
07:30:36	Reverse selected, No. 1 only	98	95	4% / --	2,940 psi	No. 2 reverse not selected; No. 2 propeller RPM decaying
07:30:44	--	58	56	4% / --	2,940 psi	Mean deceleration 07:30:34 to 07:30:44 is 2.6 m/s ²
07:30:51	Aircraft departs the sealed surface	21	20	4% / --	2,940 psi	Longitudinal deceleration falls to 1.9 m/s ² in gravel
07:30:57	Aircraft stops	0	0	4% / --	2,940 psi	62 m beyond the runway 09 end, on the extended centreline
07:31:26	Both engines shut down	0	0	-- / --	0 psi	Fuel levers to cut-off

What the data shows

Both engines were at take-off torque, within 0.1% of each other, from power set until the power levers were retarded at 07:30:32. No engine parameter exceedance, caution or warning is recorded before that point. Recorded fuel flow, ITT and propeller RPM on both engines are within normal scatter for the take-off roll. There is no data supporting a power loss before the reject.

The recorded wind at the aircraft during the roll gives a headwind component of 4 to 6 kt throughout, consistent with the 07:00 UTC routine observation.

No tailwind component is recorded at any point.

The reject was initiated 3 seconds after V1 and at 118 kt, six knots above it.

Power lever movement to flight idle took 0.6 seconds and maximum braking was established 2 seconds after the decision. That response time is at the fast end of the operator's own recorded fleet distribution for rejected take-offs (fleet mean 3.2 seconds from decision to maximum braking).

Reverse was selected on No. 1 only. No. 2 propeller RPM decays from 07:30:33, consistent with the blade damage found on inspection, and the No. 2 reverse interlock did not make. The scheduled accelerate-stop figures assume symmetric reverse; the achieved mean deceleration of 2.6 m/s² is approximately 15% below the value used in the certified accelerate-stop data for this weight and runway condition.

Distance reconstruction from recorded groundspeed puts the aircraft 1,152 m along runway 09 at the moment of the reject, leaving 695 m of sealed surface.

The stopping distance actually achieved from 118 kt was 757 m.

Source: Flight Data Section report FDA-2026-0417-R1, issued 24/06/2026

METAR YTLS 180700Z 09006KT 9999 FEW040 08/M01 Q1024 RMK RF00.0/000.0

METAR YTLS 180630Z 09007KT 9999 FEW040 09/M01 Q1024 RMK RF00.0/000.0

METAR YTLS 180600Z 08008KT 9999 SCT040 11/M01 Q1023 RMK RF00.0/000.0

METAR YTLS 180800Z 09005KT 9999 FEW040 07/M02 Q1024 RMK RF00.0/000.0

Forecast held by the crew

TAF YTLS 180458Z 1806/1824

09008KT 9999 FEW040

FM181000 08006KT CAVOK

RMK T 11 08 06 05 Q 1023 1024 1024 1025

Light and moon

Tarlton Springs, 18/06/2026 (AEST, UTC+10)

Sunset 16:58

End of civil twilight 17:25

Moonrise 22:41 (waning crescent, 14% illuminated)

Source: MET-ARCH-SB512-180626, retrieved 19/06/2026

Departure in the hour after last light, the peak of kangaroo movement

Test	Result	Supporting text
Existence	Pass - extremely likely (95%)	The roll began at 17:30 local, 32 minutes after sunset and 5 minutes after the end of civil twilight, with moonrise not until 22:41 and the moon 14% illuminated. The aerodrome was in full darkness with no natural ambient light on the strip and no town lighting within sight. The investigator's site visit at the same time of day on 23/06/2026 confirmed that the unlit graded strip either side of the sealed surface is not visible from the centreline. The aerodrome's own sighting log records 38 of its 41 kangaroo entries between one hour before and two hours after last light.
Influence	Pass - likely (80%)	The occurrence happened inside the two-hour window in which this aerodrome's own records show almost all of its wildlife movement, and in the light condition in which nothing off the sealed surface can be seen.
Importance	Not tested	(none)

Investigator notes: The condition that connects the schedule change to the animals. On its own it is unremarkable; it is what makes the missing inspection matter.

Evidence attached

METAR YTLS 180700Z 09006KT 9999 FEW040 08/M01 Q1024 RMK RF00.0/000.0

METAR YTLS 180630Z 09007KT 9999 FEW040 09/M01 Q1024 RMK RF00.0/000.0

METAR YTLS 180600Z 08008KT 9999 SCT040 11/M01 Q1023 RMK RF00.0/000.0

METAR YTLS 180800Z 09005KT 9999 FEW040 07/M02 Q1024 RMK RF00.0/000.0

Forecast held by the crew

TAF YTLS 180458Z 1806/1824

09008KT 9999 FEW040

FM181000 08006KT CAVOK

RMK T 11 08 06 05 Q 1023 1024 1024 1025

Light and moon

Tarlton Springs, 18/06/2026 (AEST, UTC+10)

Sunset 16:58

End of civil twilight 17:25

Moonrise 22:41 (waning crescent, 14% illuminated)

Source: MET-ARCH-SB512-180626, retrieved 19/06/2026

46 entries. 41 of the 46

record kangaroos on or adjacent to the movement area. 38 of those 41 are timed between one hour before and two hours after last light. Eleven entries record animals on the sealed surface of runway 09/27. The log is a paper register held in the aerodrome office. It is not reported to any operator, and the aerodrome reporting officer confirmed that no airline has ever asked for it.

Source: Tarlton Springs Aerodrome wildlife sighting log, copied 23/06/2026

Runway edge lighting intensity reduced the crew's ability to detect the animals

Test	Result	Supporting text
Existence	Inconclusive - about as likely as not (45%)	For: the pilot in command recalled the lighting as 'brighter than I'd want', a corridor of light with everything outside it black, and said anything off the seal could not be seen coming. A photometric check seven days later, on 25/06/2026, put four northern edge lights between the 900 m and 1,200 m markers at 118% to 131% of nominal for stage 3, in the section the animals crossed, and the aerodrome's maintenance programme checks lamp serviceability without ever measuring intensity. The intensity deviation is settled; the step from that reading to a reduced ability to see an animal on the runway 09 surface is an inference, and the inference is what this object tests.
Influence	Not tested	(none)
Importance	Not tested	(none)

Investigator notes: Left in draft on purpose. Both accounts are honest and they do not agree, the one measurement shows a deviation but not an effect, and there is no way to close it with what is available. Say so in the report rather than picking a side.

Evidence attached

07:29:04 PIC Saltbush five one two, ready runway zero nine.

07:29:10 TWR Saltbush five one two, wind zero nine zero at six, runway zero nine, cleared for take-off.

07:29:14 PIC Cleared for take-off runway zero nine, Saltbush five one two.

07:29:22 PIC Right, clearance is in, lights are on. Your aeroplane, your take-off.

07:29:24 FO My aeroplane, my take-off. Line up checks?

07:29:27 PIC Line up checks complete. Transponder's on.

07:29:41 FO OK. Setting power.

07:30:02 FO Power set.

07:30:04 PIC Power set, both engines good.

07:30:15 PIC Eighty knots.

07:30:16 FO Checked.

07:30:29 PIC V one.

07:30:31 FO Roos - roos, roos, on the left, they're on the -

07:30:32 PIC Stopping. I have control.

07:30:32 FO You have control.

07:30:33 [Sound consistent with a heavy impact. Duration approximately 0.4 seconds. Airframe rumble follows.]

07:30:34 FO (unintelligible) took one on the prop.

07:30:36 PIC Reverse - number two's not coming. Max brakes, max brakes.

07:30:41 FO Speed's coming down. Sixty ... fifty ...

07:30:47 PIC We're not stopping on it.

07:30:49 FO End's coming up.

07:30:51 [Sound of surface change. Increased low-frequency vibration.]

07:30:57 [Vibration ceases.]

07:30:59 PIC OK. We're stopped. Everyone alright?

07:31:01 FO Yeah. Yeah, good.

07:31:04 PIC Right. Parking brake set. Engines - both to cut-off, no fire indications, nothing on the board. I'm not evacuating.

07:31:12 FO Agreed, nothing showing.

07:31:26 [Sound consistent with both fuel levers being moved.]

07:31:34 PIC Cabin, flight deck, this is the captain, remain seated, remain seated. We have stopped on the aerodrome, there is no fire, stand by.

07:31:48 TWR Saltbush five one two, Tarlton Springs, are you able to advise your situation?

07:31:54 PIC Tarlton, Saltbush five one two, we've rejected, we've gone off the end of zero nine, we're stopped in the gravel. Forty four passengers, three crew, no injuries reported, no fire. We're going to get everyone off by the airstair. Request you activate the emergency plan and get a vehicle to us.

07:32:10 TWR Saltbush five one two, copied, activating the aerodrome emergency plan, vehicle on the way. Confirm no assistance required for injuries?

07:32:19 PIC Negative at this stage, we'll advise.

07:32:31 PIC [To first officer] Did you see them coming, or did they just turn up?

07:32:35 FO They came out of the dark on the left of the strip, crossed the centreline. I had them for - I don't know, two seconds. Maybe less. There was nothing to see and then there were two of them.

07:32:47 PIC I never had them. I was heads-in on the engine page after V one.

07:33:02 PIC Cabin, flight deck. We're going to disembark by the forward door onto the ground. It's dark and it's gravel out there, so nice and slow, no rush, no evacuation.

07:34:40 [Sound consistent with the forward airstair operating.]

Source: CVR-2026-0417, transcribed 25/06/2026, verified by a second investigator

Carried out at the investigator's request. Four runway 09/27 edge lights on the northern side between the 900 m and 1,200 m markers measured between 118% and 131% of the nominal intensity for stage 3. The remainder were within tolerance. The aerodrome operator's own maintenance programme checks lamp serviceability but does not measure intensity.

Source: Photometric check, Tarlton Springs runway 09/27 edge lighting, 25/06/2026

Fifth sector, 11 hours into duty, 40 minutes behind schedule

Test	Result	Supporting text
Existence	Pass - virtually certain (100%)	Sign-on 06:30 local, five sectors, the fifth added that morning when the rostered crew reported unfit. The occurrence was at 17:30 local, 11 hours 00 minutes into a duty planned to finish at 18:35, giving 12 hours 05 against a limit of 12 hours 30 for five sectors from that sign-on. The aircraft departed 40 minutes behind schedule. Both crew described themselves in interview as tired but not struggling; the captain said he would have declined a further sector and that 'for the two seconds that mattered, I was there'.
Influence	Fail - unlikely (25%)	A late fifth sector at the end of a long duty into an aerodrome with no alternate at night is a routine exposure on this schedule, not a one-off.
Importance	Pass - likely (85%)	It failed Influence for this occurrence and remains a finding worth acting on. The evening Tarlton Springs rotation routinely places the last sector inside the last hour of a 12-hour duty, at an aerodrome with no company engineering, no alternate within 200 nm at night, and no crewing resilience, so a single unfit crew member turns a four-sector day into a five-sector one on the morning of operation. Nothing about that changes because it did not bite here.

Investigator notes: The tempting object. It exists, it reads as a cause, and the evidence will not carry it for this occurrence. Keeping it as an other risk factor is the honest answer and it is still going in the report. No arrow, deliberately: Influence failed at 25%, so every mechanism fatigue would have had to work through is contradicted by the recorded data, and an arrow would assert a causal path this record does not have. It did not contribute to the overrun; it is a finding about the schedule.

Evidence attached

UTC	Event	IAS	Groundspeed	Torque 1 / 2	Brake pressure	Notes
07:30:02	Power set, roll commences	0	0	98.4% / 98.5%	0 psi	Both power levers advanced together
07:30:15	80 kt cross-check	80	77	98.4% / 98.5%	0 psi	No caution or warning active
07:30:29	V1	112	109	98.4% / 98.5%	0 psi	Recorded aural "V1" on CVR
07:30:31	First officer's sighting call	116	113	98.4% / 98.5%	0 psi	No control input yet
07:30:32	Reject initiated	118	115	98.4% / 98.5%	0 psi	Both power levers to flight idle within 0.6 s
07:30:33	Impact	114	111	12% / 9%	2,180 psi	Lateral acceleration spike 0.31 g right; vertical spike 1.48 g
07:30:34	Maximum braking established	109	106	6% / 6%	2,940 psi	Anti-skid active on all four wheels
07:30:36	Reverse selected, No. 1 only	98	95	4% / --	2,940 psi	No. 2 reverse not selected; No. 2 propeller RPM decaying
07:30:44	--	58	56	4% / --	2,940 psi	Mean deceleration 07:30:34 to 07:30:44 is 2.6 m/s ²
07:30:51	Aircraft departs the sealed surface	21	20	4% / --	2,940 psi	Longitudinal deceleration falls to 1.9 m/s ² in gravel

07:30:57	Aircraft stops	0	0	4% / --	2,940 psi	62 m beyond the runway 09 end, on the extended centreline
07:31:26	Both engines shut down	0	0	-- / --	0 psi	Fuel levers to cut-off

What the data shows

Both engines were at take-off torque, within 0.1% of each other, from power set until the power levers were retarded at 07:30:32. No engine parameter exceedance, caution or warning is recorded before that point. Recorded fuel flow, ITT and propeller RPM on both engines are within normal scatter for the take-off roll. There is no data supporting a power loss before the reject.

The recorded wind at the aircraft during the roll gives a headwind component of 4 to 6 kt throughout, consistent with the 07:00 UTC routine observation.

No tailwind component is recorded at any point.

The reject was initiated 3 seconds after V1 and at 118 kt, six knots above it.

Power lever movement to flight idle took 0.6 seconds and maximum braking was established 2 seconds after the decision. That response time is at the fast end of the operator's own recorded fleet distribution for rejected take-offs (fleet mean 3.2 seconds from decision to maximum braking).

Reverse was selected on No. 1 only. No. 2 propeller RPM decays from 07:30:33, consistent with the blade damage found on inspection, and the No. 2 reverse interlock did not make. The scheduled accelerate-stop figures assume symmetric reverse; the achieved mean deceleration of 2.6 m/s² is approximately 15% below the value used in the certified accelerate-stop data for this weight and runway condition.

Distance reconstruction from recorded groundspeed puts the aircraft 1,152 m along runway 09 at the moment of the reject, leaving 695 m of sealed surface.

The stopping distance actually achieved from 118 kt was 757 m.

Source: Flight Data Section report FDA-2026-0417-R1, issued 24/06/2026

Pilot in command. Total 8,410 hours, 2,140 on type, in command 3 years 1 month. Last cyclic simulator 11/03/2026, all items satisfactory. Last line check 02/12/2025, satisfactory. No prior occurrence history. Duty and rest in the 7 days before the occurrence within the company's limits with no variations applied.

First officer. Total 1,655 hours, 405 on type, 14 months with the company. Last cyclic simulator 28/04/2026, all items satisfactory. Two prior sectors into Tarlton Springs, both in daylight, on 09/01/2026 and 17/02/2026.

Duty record, 18/06/2026. Sign-on 06:30 local. Five sectors. The occurrence occurred at 17:30 local, 11 hours 00 minutes into a duty planned to finish at 18:35 local, giving a 12 hour 05 minute duty. The maximum permitted duty for five sectors with this sign-on is 12 hours 30 minutes. The fifth sector was added on the morning of 18/06/2026 when the rostered crew reported unfit. The schedule was 40 minutes late at Tarlton Springs departure.

Source: Crew records and duty record, 18/06/2026

Individual Actions

No runway inspection requested before the first departure after last light

Test	Result	Supporting text
Existence	Pass - virtually certain (100%)	Neither the CVR nor the tower's recorded audio contains any request for a runway inspection, any offer of one, or any mention of wildlife at any point before departure. The aerodrome's daily inspection log for 18/06/2026 records one inspection, at 07:35 local before the first arrival, result 'movement area clear', with none afterwards and none requested. Aerodrome Manual 6.3 provides for inspections outside the scheduled time on the request of the reporting officer or a pilot in command, and the reporting officer confirmed one would have been carried out if asked. Both crew stated they did not consider it; the captain had expected Tarlton Springs to be on the Appendix 8A list and was surprised to find it was not.
Influence	Pass - likely (85%)	The one action that would have put a vehicle on the runway in the 20 minutes before the roll was available, costs nothing, and was never on either crew member's mind because no document put it there.
Importance	Not tested	(none)

Investigator notes: Careful with the wording of this one in the report. It is an omission with no owner: nothing asked either crew member to consider it, and the aerodrome would have done it if asked.

Evidence attached

07:29:04 PIC Saltbush five one two, ready runway zero nine.

07:29:10 TWR Saltbush five one two, wind zero nine zero at six, runway zero nine, cleared for take-off.

07:29:14 PIC Cleared for take-off runway zero nine, Saltbush five one two.

07:29:22 PIC Right, clearance is in, lights are on. Your aeroplane, your take-off.

07:29:24 FO My aeroplane, my take-off. Line up checks?

07:29:27 PIC Line up checks complete. Transponder's on.

07:29:41 FO OK. Setting power.

07:30:02 FO Power set.

07:30:04 PIC Power set, both engines good.

07:30:15 PIC Eighty knots.

07:30:16 FO Checked.

07:30:29 PIC V one.

07:30:31 FO Roos - roos, roos, on the left, they're on the -

07:30:32 PIC Stopping. I have control.

07:30:32 FO You have control.

07:30:33 [Sound consistent with a heavy impact. Duration approximately 0.4

seconds. Airframe rumble follows.]

07:30:34 FO (unintelligible) took one on the prop.

07:30:36 PIC Reverse - number two's not coming. Max brakes, max brakes.

07:30:41 FO Speed's coming down. Sixty ... fifty ...

07:30:47 PIC We're not stopping on it.

07:30:49 FO End's coming up.

07:30:51 [Sound of surface change. Increased low-frequency vibration.]

07:30:57 [Vibration ceases.]

07:30:59 PIC OK. We're stopped. Everyone alright?

07:31:01 FO Yeah. Yeah, good.

07:31:04 PIC Right. Parking brake set. Engines - both to cut-off, no fire indications, nothing on the board. I'm not evacuating.

07:31:12 FO Agreed, nothing showing.

07:31:26 [Sound consistent with both fuel levers being moved.]

07:31:34 PIC Cabin, flight deck, this is the captain, remain seated, remain seated. We have stopped on the aerodrome, there is no fire, stand by.

07:31:48 TWR Saltbush five one two, Tarlton Springs, are you able to advise your situation?

07:31:54 PIC Tarlton, Saltbush five one two, we've rejected, we've gone off the end of zero nine, we're stopped in the gravel. Forty four passengers, three crew, no injuries reported, no fire.

We're going to get everyone off by the airstair. Request you activate the emergency plan and get a vehicle to us.

07:32:10 TWR Saltbush five one two, copied, activating the aerodrome emergency plan, vehicle on the way. Confirm no assistance required for injuries?

07:32:19 PIC Negative at this stage, we'll advise.

07:32:31 PIC [To first officer] Did you see them coming, or did they just turn up?

07:32:35 FO They came out of the dark on the left of the strip, crossed the centreline. I had them for - I don't know, two seconds. Maybe less. There was nothing to see and then there were two of them.

07:32:47 PIC I never had them. I was heads-in on the engine page after V one.

07:33:02 PIC Cabin, flight deck. We're going to disembark by the forward

door onto the ground. It's dark and it's gravel out there, so nice and slow, no rush, no evacuation.

07:34:40 [Sound consistent with the forward airstair operating.]

Source: CVR-2026-0417, transcribed 25/06/2026, verified by a second investigator

8.14.1 Wildlife presents a hazard to aircraft operations, particularly at regional aerodromes and particularly around dawn and dusk.

8.14.2 At an aerodrome listed in Appendix 8A, the pilot in command shall request a runway inspection from the aerodrome operator prior to the first departure after last light, and shall not commence the take-off roll until the result of that inspection has been passed.

8.14.3 At all other aerodromes, no specific action is required. Crews should remain alert to the possibility of wildlife on the movement area.

8.14.4 A wildlife strike shall be reported on an air safety report. Where a strike occurs, the Safety Department will review the aerodrome's status against the criteria in Appendix 8A.

Appendix 8A, Wildlife-active aerodromes. An aerodrome is listed in this appendix where a wildlife strike involving a company aircraft has been reported at that aerodrome within the preceding 24 months. The list is reviewed annually by the Safety Department.

Appendix 8A as current on 18/06/2026 lists nine aerodromes. Tarlton Springs is not among them, and has never been listed. The list was last reviewed on 03/03/2025.

The gap. The only route onto the list is a reported strike. A sighting cannot put an aerodrome on the list, and nothing in Part A asks a crew to report a sighting. Tarlton Springs has had no company strike, so it cannot be listed; because it is not listed, no inspection is requested; because no inspection is requested, nothing is looked for. The control is triggered by the outcome it exists to prevent.

Source: Saltbush Air Operations Manual Part A section 8.14, revision 9, 04/2024
46 entries. 41 of the 46

record kangaroos on or adjacent to the movement area. 38 of those 41 are timed between one hour before and two hours after last light. Eleven entries record animals on the sealed surface of runway 09/27. The log is a paper register held in the aerodrome office. It is not reported to any operator, and the aerodrome reporting officer confirmed that no airline has ever asked for it.

Source: Tarlton Springs Aerodrome wildlife sighting log, copied 23/06/2026

Take-off rejected 6 kt above V1

Test	Result	Supporting text
Existence	Pass - virtually certain (100%)	The FDR records V1 called at 112 kt at 07:30:29 and both power levers retarded to flight idle from 07:30:32 at 118 kt, three seconds and six knots past V1. Part B 4.6.4 states the take-off shall be continued at or above V1. The CVR records the whole decision in one exchange: the first officer's sighting call at 07:30:31, 'Stopping, I have control' at 07:30:32. The captain's own account, in interview and in his air safety report, is that he judged rotating into two large animals would leave him airborne at night with unassessable damage over country with no usable option, and that he took the risk whose end he could see. He states plainly that it was not by the book.
Influence	Pass - extremely likely (95%)	Rejecting at 118 kt rather than continuing is the decision that put the aircraft into a stop it did not have the runway for. The scheduled accelerate-stop distance of 1,780 m assumes the reject begins at or before V1.
Importance	Not tested	(none)

Investigator notes: Do not let this one carry the report. It is a contributing factor and it is downstream of the training object and the animals being there at all.

Evidence attached

UTC	Event	IAS	Groundspeed	Torque 1 / 2	Brake pressure	Notes
07:30:02	Power set, roll commences	0	0	98.4% / 98.5%	0 psi	Both power levers advanced together
07:30:15	80 kt cross-check	80	77	98.4% / 98.5%	0 psi	No caution or warning active
07:30:29	V1	112	109	98.4% / 98.5%	0 psi	Recorded aural "V1" on CVR
07:30:31	First officer's sighting call	116	113	98.4% / 98.5%	0 psi	No control input yet
07:30:32	Reject initiated	118	115	98.4% / 98.5%	0 psi	Both power levers to flight idle within 0.6 s
07:30:33	Impact	114	111	12% / 9%	2,180 psi	Lateral acceleration spike 0.31 g right; vertical spike 1.48 g
07:30:34	Maximum braking established	109	106	6% / 6%	2,940 psi	Anti-skid active on all four wheels
07:30:36	Reverse selected, No. 1 only	98	95	4% / --	2,940 psi	No. 2 reverse not selected; No. 2 propeller RPM decaying
07:30:44	--	58	56	4% / --	2,940 psi	Mean deceleration 07:30:34 to 07:30:44 is 2.6 m/s ²
07:30:51	Aircraft departs the sealed surface	21	20	4% / --	2,940 psi	Longitudinal deceleration falls to 1.9 m/s ² in gravel
07:30:57	Aircraft stops	0	0	4% / --	2,940 psi	62 m beyond the runway 09 end, on the extended centreline
07:31:26	Both engines shut down	0	0	-- / --	0 psi	Fuel levers to cut-off

What the data shows

Both engines were at take-off torque, within 0.1% of each other, from power

set until the power levers were retarded at 07:30:32. No engine parameter exceedance, caution or warning is recorded before that point. Recorded fuel flow, ITT and propeller RPM on both engines are within normal scatter for the take-off roll. There is no data supporting a power loss before the reject.

The recorded wind at the aircraft during the roll gives a headwind component of 4 to 6 kt throughout, consistent with the 07:00 UTC routine observation.

No tailwind component is recorded at any point.

The reject was initiated 3 seconds after V1 and at 118 kt, six knots above it.

Power lever movement to flight idle took 0.6 seconds and maximum braking was established 2 seconds after the decision. That response time is at the fast end of the operator's own recorded fleet distribution for rejected take-offs (fleet mean 3.2 seconds from decision to maximum braking).

Reverse was selected on No. 1 only. No. 2 propeller RPM decays from 07:30:33, consistent with the blade damage found on inspection, and the No. 2 reverse interlock did not make. The scheduled accelerate-stop figures assume symmetric reverse; the achieved mean deceleration of 2.6 m/s² is approximately 15% below the value used in the certified accelerate-stop data for this weight and runway condition.

Distance reconstruction from recorded groundspeed puts the aircraft 1,152 m along runway 09 at the moment of the reject, leaving 695 m of sealed surface.

The stopping distance actually achieved from 118 kt was 757 m.

Source: Flight Data Section report FDA-2026-0417-R1, issued 24/06/2026

07:29:04 PIC Saltbush five one two, ready runway zero nine.

07:29:10 TWR Saltbush five one two, wind zero nine zero at six, runway zero nine, cleared for take-off.

07:29:14 PIC Cleared for take-off runway zero nine, Saltbush five one two.

07:29:22 PIC Right, clearance is in, lights are on. Your aeroplane, your take-off.

07:29:24 FO My aeroplane, my take-off. Line up checks?

07:29:27 PIC Line up checks complete. Transponder's on.

07:29:41 FO OK. Setting power.

07:30:02 FO Power set.

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07:30:31 FO Roos - roos, roos, on the left, they're on the -

07:30:32 PIC Stopping. I have control.

07:30:32 FO You have control.

07:30:33 [Sound consistent with a heavy impact. Duration approximately 0.4 seconds. Airframe rumble follows.]

07:30:34 FO (unintelligible) took one on the prop.

07:30:36 PIC Reverse - number two's not coming. Max brakes, max brakes.

07:30:41 FO Speed's coming down. Sixty ... fifty ...

07:30:47 PIC We're not stopping on it.

07:30:49 FO End's coming up.

07:30:51 [Sound of surface change. Increased low-frequency vibration.]

07:30:57 [Vibration ceases.]

07:30:59 PIC OK. We're stopped. Everyone alright?

07:31:01 FO Yeah. Yeah, good.

07:31:04 PIC Right. Parking brake set. Engines - both to cut-off, no fire indications, nothing on the board. I'm not evacuating.

07:31:12 FO Agreed, nothing showing.

07:31:26 [Sound consistent with both fuel levers being moved.]

07:31:34 PIC Cabin, flight deck, this is the captain, remain seated, remain seated. We have stopped on the aerodrome, there is no fire, stand by.

07:31:48 TWR Saltbush five one two, Tarlton Springs, are you able to advise your situation?

07:31:54 PIC Tarlton, Saltbush five one two, we've rejected, we've gone off the end of zero nine, we're stopped in the gravel. Forty four passengers, three crew, no injuries reported, no fire. We're going to get everyone off by the airstair. Request you activate the emergency plan and get a vehicle to us.

07:32:10 TWR Saltbush five one two, copied, activating the aerodrome emergency plan, vehicle on the way. Confirm no assistance required for injuries?

07:32:19 PIC Negative at this stage, we'll advise.

07:32:31 PIC [To first officer] Did you see them coming, or did they just turn up?

07:32:35 FO They came out of the dark on the left of the strip, crossed the centreline. I had them for - I don't know, two seconds. Maybe less. There was nothing to see and then there were two of them.

07:32:47 PIC I never had them. I was heads-in on the engine page after V one.

07:33:02 PIC Cabin, flight deck. We're going to disembark by the forward door onto the ground. It's dark and it's gravel out there, so nice and slow, no rush, no evacuation.

07:34:40 [Sound consistent with the forward airstair operating.]

Source: CVR-2026-0417, transcribed 25/06/2026, verified by a second investigator

4.6.2 Below 80 kt the take-off shall be rejected for any malfunction or condition that in the judgement of the pilot in command makes the aircraft unfit to fly.

4.6.3 Between 80 kt and V1 the take-off shall be rejected only for: fire or fire warning; engine failure; a configuration warning; loss of directional control; or the pilot in command's judgement that the aircraft is unsafe or unable to fly.

4.6.4 At or above V1 the take-off shall be continued. The decision to continue is made before V1 is reached.

Section 4.6 contains no guidance on an obstruction seen on the runway, at any speed, and no amplification of 4.6.4 for a case in which continuing is itself hazardous.

Source: Saltbush Air Operations Manual Part B section 4.6, revision 14, 09/2025

Maximum braking and No. 1 reverse applied without delay

Test	Result	Supporting text
Existence	Pass - virtually certain (100%)	Power levers reached flight idle 0.6 seconds after the decision and maximum braking was established at 07:30:34, 2 seconds after it, against a recorded fleet mean of 3.2 seconds from decision to maximum braking. Anti-skid was active on all four wheels throughout and brake energy reached 84% of the certified maximum without a brake fire. Reverse was selected on No. 1 within 4 seconds; No. 2 was unavailable because its propeller had been destroyed in the strike. The first officer made the speed callouts and the runway-end call. The aircraft left the seal at 21 kt, stopped 62 m into a 90 m RESA, upright, on the extended centreline, and was disembarked without an evacuation.
Influence	Pass - very likely (90%)	The overrun was 62 m into a prepared surface at low speed rather than a departure from the RESA at speed. The margin between those two outcomes is the two seconds and the asymmetric braking technique.
Importance	Not tested	(none)

Investigator notes: Recorded as a positive event because it is one. The same investigation that finds the reject was outside procedure should say that what followed it was executed better than the fleet average.

Evidence attached

UTC	Event	IAS	Groundspeed	Torque 1 / 2	Brake pressure	Notes
07:30:02	Power set, roll commences	0	0	98.4% / 98.5%	0 psi	Both power levers advanced together
07:30:15	80 kt cross-check	80	77	98.4% / 98.5%	0 psi	No caution or warning active
07:30:29	V1	112	109	98.4% / 98.5%	0 psi	Recorded aural "V1" on CVR
07:30:31	First officer's sighting call	116	113	98.4% / 98.5%	0 psi	No control input yet
07:30:32	Reject initiated	118	115	98.4% / 98.5%	0 psi	Both power levers to flight idle within 0.6 s
07:30:33	Impact	114	111	12% / 9%	2,180 psi	Lateral acceleration spike 0.31 g right; vertical spike 1.48 g
07:30:34	Maximum braking established	109	106	6% / 6%	2,940 psi	Anti-skid active on all four wheels
07:30:36	Reverse selected, No. 1 only	98	95	4% / --	2,940 psi	No. 2 reverse not selected; No. 2 propeller RPM decaying
07:30:44	--	58	56	4% / --	2,940 psi	Mean deceleration 07:30:34 to 07:30:44 is 2.6 m/s ²
07:30:51	Aircraft departs the sealed surface	21	20	4% / --	2,940 psi	Longitudinal deceleration falls to 1.9 m/s ² in gravel
07:30:57	Aircraft stops	0	0	4% / --	2,940 psi	62 m beyond the runway 09 end, on the extended centreline
07:31:26	Both engines shut down	0	0	-- / --	0 psi	Fuel levers to cut-off

What the data shows

Both engines were at take-off torque, within 0.1% of each other, from power

set until the power levers were retarded at 07:30:32. No engine parameter exceedance, caution or warning is recorded before that point. Recorded fuel flow, ITT and propeller RPM on both engines are within normal scatter for the take-off roll. There is no data supporting a power loss before the reject.

The recorded wind at the aircraft during the roll gives a headwind component of 4 to 6 kt throughout, consistent with the 07:00 UTC routine observation.

No tailwind component is recorded at any point.

The reject was initiated 3 seconds after V1 and at 118 kt, six knots above it.

Power lever movement to flight idle took 0.6 seconds and maximum braking was established 2 seconds after the decision. That response time is at the fast end of the operator's own recorded fleet distribution for rejected take-offs (fleet mean 3.2 seconds from decision to maximum braking).

Reverse was selected on No. 1 only. No. 2 propeller RPM decays from 07:30:33, consistent with the blade damage found on inspection, and the No. 2 reverse interlock did not make. The scheduled accelerate-stop figures assume symmetric reverse; the achieved mean deceleration of 2.6 m/s² is approximately 15% below the value used in the certified accelerate-stop data for this weight and runway condition.

Distance reconstruction from recorded groundspeed puts the aircraft 1,152 m along runway 09 at the moment of the reject, leaving 695 m of sealed surface.

The stopping distance actually achieved from 118 kt was 757 m.

Source: Flight Data Section report FDA-2026-0417-R1, issued 24/06/2026

- No. 2 propeller: three blades damaged beyond repairable limits, leading edge deformation and one blade tip separation. Animal tissue recovered from two blade roots. Engine subject to a shock-load inspection; borescope of both engines found no pre-existing damage and no evidence of any in-flight or on-ground power anomaly before the impact.
- Right main landing gear: both tyres deflated through the fuse plugs, consistent with brake energy. Gear doors distorted by impact. Gear structure within limits on detailed inspection.
- Brakes: brake energy absorbed calculated at 84% of the certified maximum for the weight and speed. No brake fire, no hydraulic leak. Anti-skid system tested serviceable after the occurrence.
- Lower right fuselage skin: abrasion and two punctures consistent with propeller debris.
- Runway and RESA survey: two animal carcasses recovered, one at 1,188 m and one at 1,221 m from the runway 09 threshold. Continuous tyre marks from

1,160 m to the runway end. Nose wheel rut in the RESA to 62 m beyond the end, on the extended centreline.

Source: Engineering inspection report, VH-SBQ, 19/06/2026 to 21/06/2026

Occurrence Events

No. 2 engine power loss during the take-off roll

Test	Result	Supporting text
Existence	Fail - very unlikely (5%)	Hypothesis tested because a reject at 118 kt is most commonly an engine event. It fails. The FDR records both engines at 98.4% and 98.5% torque, within 0.1% of each other, from power set at 07:30:02 until the power levers were retarded at 07:30:32, with fuel flow, ITT and propeller RPM on both engines inside normal scatter throughout. No caution, warning or exceedance is recorded at any point before the reject. The CVR contains no engine-related call: the captain's first words are 'Stopping, I have control' one second after the first officer's sighting call. The borescope of both engines found no pre-existing damage. The No. 2 propeller damage and the RPM decay both begin at 07:30:33, after the reject and at the moment of impact.
Influence	Not tested	(none)
Importance	Not tested	(none)

Investigator notes: The first hypothesis anyone offers for a high-speed reject, and the first one the data kills. Kept on the map because the report should show it was tested, not assumed away. No arrow, deliberately: Existence failed, so there is nothing to draw an arrow from. The No. 2 power loss did not contribute to the reject because it had not happened yet; it begins at impact, one second after the reject.

Evidence attached

UTC	Event	IAS	Groundspeed	Torque 1 / 2	Brake pressure	Notes
07:30:02	Power set, roll commences	0	0	98.4% / 98.5%	0 psi	Both power levers advanced together
07:30:15	80 kt cross-check	80	77	98.4% / 98.5%	0 psi	No caution or warning active
07:30:29	V1	112	109	98.4% / 98.5%	0 psi	Recorded aural "V1" on CVR
07:30:31	First officer's sighting call	116	113	98.4% / 98.5%	0 psi	No control input yet
07:30:32	Reject initiated	118	115	98.4% / 98.5%	0 psi	Both power levers to flight idle within 0.6 s
07:30:33	Impact	114	111	12% / 9%	2,180 psi	Lateral acceleration spike 0.31 g right; vertical spike 1.48 g
07:30:34	Maximum braking established	109	106	6% / 6%	2,940 psi	Anti-skid active on all four wheels
07:30:36	Reverse selected, No. 1 only	98	95	4% / --	2,940 psi	No. 2 reverse not selected; No. 2 propeller RPM decaying
07:30:44	--	58	56	4% / --	2,940 psi	Mean deceleration 07:30:34 to 07:30:44 is 2.6 m/s ²
07:30:51	Aircraft departs the sealed surface	21	20	4% / --	2,940 psi	Longitudinal deceleration falls to 1.9 m/s ² in gravel
07:30:57	Aircraft stops	0	0	4% / --	2,940 psi	62 m beyond the runway 09 end, on the extended centreline

07:31:26	Both engines shut down	0	0	-- / --	0 psi	Fuel levers to cut-off
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What the data shows

Both engines were at take-off torque, within 0.1% of each other, from power set until the power levers were retarded at 07:30:32. No engine parameter exceedance, caution or warning is recorded before that point. Recorded fuel flow, ITT and propeller RPM on both engines are within normal scatter for the take-off roll. There is no data supporting a power loss before the reject.

The recorded wind at the aircraft during the roll gives a headwind component of 4 to 6 kt throughout, consistent with the 07:00 UTC routine observation.

No tailwind component is recorded at any point.

The reject was initiated 3 seconds after V1 and at 118 kt, six knots above it.

Power lever movement to flight idle took 0.6 seconds and maximum braking was established 2 seconds after the decision. That response time is at the fast end of the operator's own recorded fleet distribution for rejected take-offs (fleet mean 3.2 seconds from decision to maximum braking).

Reverse was selected on No. 1 only. No. 2 propeller RPM decays from 07:30:33, consistent with the blade damage found on inspection, and the No. 2 reverse interlock did not make. The scheduled accelerate-stop figures assume symmetric reverse; the achieved mean deceleration of 2.6 m/s² is approximately 15% below the value used in the certified accelerate-stop data for this weight and runway condition.

Distance reconstruction from recorded groundspeed puts the aircraft 1,152 m along runway 09 at the moment of the reject, leaving 695 m of sealed surface.

The stopping distance actually achieved from 118 kt was 757 m.

Source: Flight Data Section report FDA-2026-0417-R1, issued 24/06/2026

- No. 2 propeller: three blades damaged beyond repairable limits, leading edge deformation and one blade tip separation. Animal tissue recovered from two blade roots. Engine subject to a shock-load inspection; borescope of both engines found no pre-existing damage and no evidence of any in-flight or on-ground power anomaly before the impact.
- Right main landing gear: both tyres deflated through the fuse plugs, consistent with brake energy. Gear doors distorted by impact. Gear structure within limits on detailed inspection.
- Brakes: brake energy absorbed calculated at 84% of the certified maximum for the weight and speed. No brake fire, no hydraulic leak. Anti-skid system tested serviceable after the occurrence.

- Lower right fuselage skin: abrasion and two punctures consistent with propeller debris.
- Runway and RESA survey: two animal carcasses recovered, one at 1,188 m and one at 1,221 m from the runway 09 threshold. Continuous tyre marks from 1,160 m to the runway end. Nose wheel rut in the RESA to 62 m beyond the end, on the extended centreline.

Source: Engineering inspection report, VH-SBQ, 19/06/2026 to 21/06/2026

Two kangaroos entered the runway strip and moved onto the sealed surface

Test	Result	Supporting text
Existence	Pass - virtually certain (100%)	Two adult kangaroos were recovered from runway 09 at 1,188 m and 1,221 m from the threshold, on the northern half. The first officer saw them from the flight deck crossing from the northern side to the south, estimating a range of around 200 m at first sight, and called them at 07:30:31. The animal pad through the failed northern fence is at the 1,050 m mark, 138 m before the first carcass on the same side. Eleven entries in the aerodrome's sighting log record kangaroos on the sealed surface of runway 09/27, all around last light. No inspection had been carried out since 07:35 local that morning.
Influence	Pass - extremely likely (95%)	Nothing else on this map matters without this. It is also the only object on the map that a single \$41,000 works item would have addressed directly.
Importance	Not tested	(none)

Investigator notes: The event everything upstream is about. Note the carcass positions against the fence pad: the geometry does the arguing here, not the interviews.

Evidence attached

07:29:04 PIC Saltbush five one two, ready runway zero nine.

07:29:10 TWR Saltbush five one two, wind zero nine zero at six, runway zero nine, cleared for take-off.

07:29:14 PIC Cleared for take-off runway zero nine, Saltbush five one two.

07:29:22 PIC Right, clearance is in, lights are on. Your aeroplane, your take-off.

07:29:24 FO My aeroplane, my take-off. Line up checks?

07:29:27 PIC Line up checks complete. Transponder's on.

07:29:41 FO OK. Setting power.

07:30:02 FO Power set.

07:30:04 PIC Power set, both engines good.

07:30:15 PIC Eighty knots.

07:30:16 FO Checked.

07:30:29 PIC V one.

07:30:31 FO Roos - roos, roos, on the left, they're on the -

07:30:32 PIC Stopping. I have control.

07:30:32 FO You have control.

07:30:33 [Sound consistent with a heavy impact. Duration approximately 0.4 seconds. Airframe rumble follows.]

07:30:34 FO (unintelligible) took one on the prop.

07:30:36 PIC Reverse - number two's not coming. Max brakes, max brakes.

07:30:41 FO Speed's coming down. Sixty ... fifty ...

07:30:47 PIC We're not stopping on it.

07:30:49 FO End's coming up.

07:30:51 [Sound of surface change. Increased low-frequency vibration.]

07:30:57 [Vibration ceases.]

07:30:59 PIC OK. We're stopped. Everyone alright?

07:31:01 FO Yeah. Yeah, good.

07:31:04 PIC Right. Parking brake set. Engines - both to cut-off, no fire indications, nothing on the board. I'm not evacuating.

07:31:12 FO Agreed, nothing showing.

07:31:26 [Sound consistent with both fuel levers being moved.]

07:31:34 PIC Cabin, flight deck, this is the captain, remain seated, remain seated. We have stopped on the aerodrome, there is no fire, stand by.

07:31:48 TWR Saltbush five one two, Tarlton Springs, are you able to advise your situation?

07:31:54 PIC Tarlton, Saltbush five one two, we've rejected, we've gone off the end of zero nine, we're stopped in the gravel. Forty four passengers, three crew, no injuries reported, no fire. We're going to get everyone off by the airstair. Request you activate the emergency plan and get a vehicle to us.

07:32:10 TWR Saltbush five one two, copied, activating the aerodrome emergency plan, vehicle on the way. Confirm no assistance required for injuries?

07:32:19 PIC Negative at this stage, we'll advise.

07:32:31 PIC [To first officer] Did you see them coming, or did they just turn up?

07:32:35 FO They came out of the dark on the left of the strip, crossed the centreline. I had them for - I don't know, two seconds. Maybe less. There was nothing to see and then there were two of them.

07:32:47 PIC I never had them. I was heads-in on the engine page after V one.

07:33:02 PIC Cabin, flight deck. We're going to disembark by the forward door onto the ground. It's dark and it's gravel out there, so nice and slow, no rush, no evacuation.

07:34:40 [Sound consistent with the forward airstair operating.]

Source: CVR-2026-0417, transcribed 25/06/2026, verified by a second investigator

46 entries. 41 of the 46

record kangaroos on or adjacent to the movement area. 38 of those 41 are timed between one hour before and two hours after last light. Eleven entries record animals on the sealed surface of runway 09/27. The log is a paper register held in the aerodrome office. It is not reported to any operator, and the aerodrome reporting officer confirmed that no airline has ever asked for it.

Source: Tarlton Springs Aerodrome wildlife sighting log, copied 23/06/2026

Item WR-221, raised 04/2023: "Perimeter fence,

northern boundary between the 700 m and 1,300 m runway markers, four panels displaced and wire strained, animals observed passing through. Recommend replacement, estimated \$41,000." Deferred at the 2023, 2024 and 2025 budget rounds, each time with the entry "defer, unfunded, review next cycle". Item WR-263, raised 02/2026, records the same section as "deteriorating further".

A site inspection on 23/06/2026 found a continuous 140 m section of the northern fence with the bottom two wires separated from the posts and a clear animal pad through it at the 1,050 m mark.

Source: Tarlton Shire Council aerodrome works register WR-221 (04/2023), WR-263 (02/2026)

- No. 2 propeller: three blades damaged beyond repairable limits, leading edge deformation and one blade tip separation. Animal tissue recovered from two blade roots. Engine subject to a shock-load inspection; borescope of both engines found no pre-existing damage and no evidence of any in-flight or on-ground power anomaly before the impact.
- Right main landing gear: both tyres deflated through the fuse plugs, consistent with brake energy. Gear doors distorted by impact. Gear structure within limits on detailed inspection.
- Brakes: brake energy absorbed calculated at 84% of the certified maximum for the weight and speed. No brake fire, no hydraulic leak. Anti-skid system tested serviceable after the occurrence.
- Lower right fuselage skin: abrasion and two punctures consistent with propeller debris.
- Runway and RESA survey: two animal carcasses recovered, one at 1,188 m and one at 1,221 m from the runway 09 threshold. Continuous tyre marks from 1,160 m to the runway end. Nose wheel rut in the RESA to 62 m beyond the end, on the extended centreline.

Source: Engineering inspection report, VH-SBQ, 19/06/2026 to 21/06/2026

No. 2 propeller and right main gear struck two kangaroos

Test	Result	Supporting text
Existence	Pass - virtually certain (100%)	Impact at 07:30:33 at 114 kt, one second after the reject began. The FDR records a lateral acceleration spike of 0.31 g right and a vertical spike of 1.48 g. Three No. 2 propeller blades were damaged beyond repairable limits with one blade tip separated, and animal tissue was recovered from two blade roots. The right main gear doors were distorted and the lower right fuselage skin punctured by propeller debris. No. 2 propeller RPM decays from the moment of impact and the No. 2 reverse interlock did not make.
Influence	Pass - virtually certain (100%)	The strike removed one of the two reverse units two seconds into a stop with a 67 m scheduled margin, and it is why the aircraft went off the end rather than stopping on it.
Importance	Not tested	(none)

Investigator notes: The link to make sure the report draws: this is not just damage, it took away half the reverse thrust in the middle of a stop that had no margin.

Evidence attached

UTC	Event	IAS	Groundspeed	Torque 1 / 2	Brake pressure	Notes
07:30:02	Power set, roll commences	0	0	98.4% / 98.5%	0 psi	Both power levers advanced together
07:30:15	80 kt cross-check	80	77	98.4% / 98.5%	0 psi	No caution or warning active
07:30:29	V1	112	109	98.4% / 98.5%	0 psi	Recorded aural "V1" on CVR
07:30:31	First officer's sighting call	116	113	98.4% / 98.5%	0 psi	No control input yet
07:30:32	Reject initiated	118	115	98.4% / 98.5%	0 psi	Both power levers to flight idle within 0.6 s
07:30:33	Impact	114	111	12% / 9%	2,180 psi	Lateral acceleration spike 0.31 g right; vertical spike 1.48 g
07:30:34	Maximum braking established	109	106	6% / 6%	2,940 psi	Anti-skid active on all four wheels
07:30:36	Reverse selected, No. 1 only	98	95	4% / --	2,940 psi	No. 2 reverse not selected; No. 2 propeller RPM decaying
07:30:44	--	58	56	4% / --	2,940 psi	Mean deceleration 07:30:34 to 07:30:44 is 2.6 m/s ²
07:30:51	Aircraft departs the sealed surface	21	20	4% / --	2,940 psi	Longitudinal deceleration falls to 1.9 m/s ² in gravel
07:30:57	Aircraft stops	0	0	4% / --	2,940 psi	62 m beyond the runway 09 end, on the extended centreline
07:31:26	Both engines shut down	0	0	-- / --	0 psi	Fuel levers to cut-off

What the data shows

Both engines were at take-off torque, within 0.1% of each other, from power set until the power levers were retarded at 07:30:32. No engine parameter

exceedance, caution or warning is recorded before that point. Recorded fuel flow, ITT and propeller RPM on both engines are within normal scatter for the take-off roll. There is no data supporting a power loss before the reject.

The recorded wind at the aircraft during the roll gives a headwind component of 4 to 6 kt throughout, consistent with the 07:00 UTC routine observation.

No tailwind component is recorded at any point.

The reject was initiated 3 seconds after V1 and at 118 kt, six knots above it.

Power lever movement to flight idle took 0.6 seconds and maximum braking was established 2 seconds after the decision. That response time is at the fast end of the operator's own recorded fleet distribution for rejected take-offs (fleet mean 3.2 seconds from decision to maximum braking).

Reverse was selected on No. 1 only. No. 2 propeller RPM decays from 07:30:33, consistent with the blade damage found on inspection, and the No. 2 reverse interlock did not make. The scheduled accelerate-stop figures assume symmetric reverse; the achieved mean deceleration of 2.6 m/s² is approximately 15% below the value used in the certified accelerate-stop data for this weight and runway condition.

Distance reconstruction from recorded groundspeed puts the aircraft 1,152 m along runway 09 at the moment of the reject, leaving 695 m of sealed surface.

The stopping distance actually achieved from 118 kt was 757 m.

Source: Flight Data Section report FDA-2026-0417-R1, issued 24/06/2026

- No. 2 propeller: three blades damaged beyond repairable limits, leading edge deformation and one blade tip separation. Animal tissue recovered from two blade roots. Engine subject to a shock-load inspection; borescope of both engines found no pre-existing damage and no evidence of any in-flight or on-ground power anomaly before the impact.
- Right main landing gear: both tyres deflated through the fuse plugs, consistent with brake energy. Gear doors distorted by impact. Gear structure within limits on detailed inspection.
- Brakes: brake energy absorbed calculated at 84% of the certified maximum for the weight and speed. No brake fire, no hydraulic leak. Anti-skid system tested serviceable after the occurrence.
- Lower right fuselage skin: abrasion and two punctures consistent with propeller debris.
- Runway and RESA survey: two animal carcasses recovered, one at 1,188 m and one at 1,221 m from the runway 09 threshold. Continuous tyre marks from 1,160 m to the runway end. Nose wheel rut in the RESA to 62 m beyond the

end, on the extended centreline.

Source: Engineering inspection report, VH-SBQ, 19/06/2026 to 21/06/2026

Deceleration achieved below the scheduled accelerate-stop performance

Test	Result	Supporting text
Existence	Pass - virtually certain (100%)	Scheduled accelerate-stop distance required was 1,780 m against an ASDA of 1,847 m, a margin of 67 m, assuming the stop begins at or before V1 with symmetric reverse. Distance reconstruction from recorded groundspeed puts the aircraft 1,152 m along the runway at the reject, leaving 695 m of sealed surface, and the stopping distance actually achieved from 118 kt was 757 m. The mean deceleration from maximum braking to the runway end was 2.6 m/s ² , approximately 15% below the value used in the certified data for this weight and condition, with reverse available on No. 1 only.
Influence	Pass - virtually certain (100%)	62 m of overrun is the arithmetic of a stop begun 6 kt late with 15% less deceleration than the figures assume. Both contributions are needed to reach the end; neither alone does.
Importance	Not tested	(none)

Investigator notes: The convergence point. Both chains arrive here, which is why the report's analysis should treat it as one mechanism rather than two separate findings.

Evidence attached

UTC	Event	IAS	Groundspeed	Torque 1 / 2	Brake pressure	Notes
07:30:02	Power set, roll commences	0	0	98.4% / 98.5%	0 psi	Both power levers advanced together
07:30:15	80 kt cross-check	80	77	98.4% / 98.5%	0 psi	No caution or warning active
07:30:29	V1	112	109	98.4% / 98.5%	0 psi	Recorded aural "V1" on CVR
07:30:31	First officer's sighting call	116	113	98.4% / 98.5%	0 psi	No control input yet
07:30:32	Reject initiated	118	115	98.4% / 98.5%	0 psi	Both power levers to flight idle within 0.6 s
07:30:33	Impact	114	111	12% / 9%	2,180 psi	Lateral acceleration spike 0.31 g right; vertical spike 1.48 g
07:30:34	Maximum braking established	109	106	6% / 6%	2,940 psi	Anti-skid active on all four wheels
07:30:36	Reverse selected, No. 1 only	98	95	4% / --	2,940 psi	No. 2 reverse not selected; No. 2 propeller RPM decaying
07:30:44	--	58	56	4% / --	2,940 psi	Mean deceleration 07:30:34 to 07:30:44 is 2.6 m/s ²
07:30:51	Aircraft departs the sealed surface	21	20	4% / --	2,940 psi	Longitudinal deceleration falls to 1.9 m/s ² in gravel
07:30:57	Aircraft stops	0	0	4% / --	2,940 psi	62 m beyond the runway 09 end, on the extended centreline
07:31:26	Both engines shut down	0	0	-- / --	0 psi	Fuel levers to cut-off

What the data shows

Both engines were at take-off torque, within 0.1% of each other, from power

set until the power levers were retarded at 07:30:32. No engine parameter exceedance, caution or warning is recorded before that point. Recorded fuel flow, ITT and propeller RPM on both engines are within normal scatter for the take-off roll. There is no data supporting a power loss before the reject.

The recorded wind at the aircraft during the roll gives a headwind component of 4 to 6 kt throughout, consistent with the 07:00 UTC routine observation.

No tailwind component is recorded at any point.

The reject was initiated 3 seconds after V1 and at 118 kt, six knots above it.

Power lever movement to flight idle took 0.6 seconds and maximum braking was established 2 seconds after the decision. That response time is at the fast end of the operator's own recorded fleet distribution for rejected take-offs (fleet mean 3.2 seconds from decision to maximum braking).

Reverse was selected on No. 1 only. No. 2 propeller RPM decays from 07:30:33, consistent with the blade damage found on inspection, and the No. 2 reverse interlock did not make. The scheduled accelerate-stop figures assume symmetric reverse; the achieved mean deceleration of 2.6 m/s² is approximately 15% below the value used in the certified accelerate-stop data for this weight and runway condition.

Distance reconstruction from recorded groundspeed puts the aircraft 1,152 m along runway 09 at the moment of the reject, leaving 695 m of sealed surface.

The stopping distance actually achieved from 118 kt was 757 m.

Source: Flight Data Section report FDA-2026-0417-R1, issued 24/06/2026

- No. 2 propeller: three blades damaged beyond repairable limits, leading edge deformation and one blade tip separation. Animal tissue recovered from two blade roots. Engine subject to a shock-load inspection; borescope of both engines found no pre-existing damage and no evidence of any in-flight or on-ground power anomaly before the impact.
- Right main landing gear: both tyres deflated through the fuse plugs, consistent with brake energy. Gear doors distorted by impact. Gear structure within limits on detailed inspection.
- Brakes: brake energy absorbed calculated at 84% of the certified maximum for the weight and speed. No brake fire, no hydraulic leak. Anti-skid system tested serviceable after the occurrence.
- Lower right fuselage skin: abrasion and two punctures consistent with propeller debris.
- Runway and RESA survey: two animal carcasses recovered, one at 1,188 m and one at 1,221 m from the runway 09 threshold. Continuous tyre marks from

1,160 m to the runway end. Nose wheel rut in the RESA to 62 m beyond the end, on the extended centreline.

Source: Engineering inspection report, VH-SBQ, 19/06/2026 to 21/06/2026

Runway overrun onto the runway end safety area

Test	Result	Supporting text
Existence	Pass - virtually certain (100%)	Saltbush Air SB512, a 50-seat twin-turboprop registered VH-SBQ, departed the end of runway 09 at Tarlton Springs Aerodrome at approximately 21 kt at 07:30:51 UTC on 18/06/2026 and came to rest at 07:30:57, 62 m into the 90 m runway end safety area, upright and on the extended centreline, with the nose wheel in graded gravel. 44 passengers, 2 flight crew and 1 cabin crew were on board. Continuous tyre marks run from 1,160 m to the runway end and the nose wheel rut continues to the stopping point.
Influence	Pass - virtually certain (100%)	Substantial damage: three No. 2 propeller blades beyond limits, a shock-load inspection on the engine, distorted right main gear doors, both right main tyres deflated through the fuse plugs at 84% of certified brake energy, and punctures to the lower right fuselage. Nil serious injuries; one cabin crew member reported bruising during the disembarkation. Two kangaroos killed. The aerodrome closed for 4 hours 20 minutes. The aircraft stopped 28 m short of the end of the prepared safety area; beyond it the terrain is ungraded.
Importance	Not tested	(none)

Investigator notes: The critical event. Everything else on this map exists to explain it.

Evidence attached

SB512 was the last scheduled sector of the day, departing 40 minutes behind schedule. The aerodrome was in darkness; the tower was open and closed at 18:00 local as rostered.

During the take-off roll on runway 09 the first officer saw two kangaroos moving onto the runway from the northern side, past the 1,100 m point. The captain rejected the take-off at 118 kt, six knots above the calculated V1 of 112 kt. About one second later the aircraft struck both animals; the right main landing gear and the No. 2 propeller took the impact.

The aircraft did not stop on the sealed surface. It departed the end of runway 09 at approximately 21 kt and came to rest 62 m into the 90 m runway end safety area, upright, on the extended centreline, with the nose wheel in graded gravel.

The captain shut both engines down, the cabin crew member was instructed to hold, and passengers were disembarked by the forward airstair 4 minutes 8 seconds after the aircraft stopped. No evacuation was commanded and no slides were deployed. Aerodrome staff attended with a vehicle; there was no fire.

Immediate actions taken

- Aerodrome closed by the aerodrome operator pending inspection and the removal of animal remains and propeller debris.
- Aircraft quarantined at Tarlton Springs. FDR and CVR removed by the operator's engineering representative at 12:40 UTC on 18/06/2026 and couriered to the operator's flight data section.

- Both flight crew stood down from duty and drug and alcohol tested; both results negative.
- The evening Tarlton Springs service suspended by the Head of Flying Operations pending the outcome of this investigation.

Source: Notification SBA-OCC-2026-0417, raised 18/06/2026 08:14 UTC

UTC	Event	IAS	Groundspeed	Torque 1 / 2	Brake pressure	Notes
07:30:02	Power set, roll commences	0	0	98.4% / 98.5%	0 psi	Both power levers advanced together
07:30:15	80 kt cross-check	80	77	98.4% / 98.5%	0 psi	No caution or warning active
07:30:29	V1	112	109	98.4% / 98.5%	0 psi	Recorded aural "V1" on CVR
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07:30:32	Reject initiated	118	115	98.4% / 98.5%	0 psi	Both power levers to flight idle within 0.6 s
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07:30:34	Maximum braking established	109	106	6% / 6%	2,940 psi	Anti-skid active on all four wheels
07:30:36	Reverse selected, No. 1 only	98	95	4% / --	2,940 psi	No. 2 reverse not selected; No. 2 propeller RPM decaying
07:30:44	--	58	56	4% / --	2,940 psi	Mean deceleration 07:30:34 to 07:30:44 is 2.6 m/s ²
07:30:51	Aircraft departs the sealed surface	21	20	4% / --	2,940 psi	Longitudinal deceleration falls to 1.9 m/s ² in gravel
07:30:57	Aircraft stops	0	0	4% / --	2,940 psi	62 m beyond the runway 09 end, on the extended centreline
07:31:26	Both engines shut down	0	0	-- / --	0 psi	Fuel levers to cut-off

What the data shows

Both engines were at take-off torque, within 0.1% of each other, from power set until the power levers were retarded at 07:30:32. No engine parameter exceedance, caution or warning is recorded before that point. Recorded fuel flow, ITT and propeller RPM on both engines are within normal scatter for the take-off roll. There is no data supporting a power loss before the reject.

The recorded wind at the aircraft during the roll gives a headwind component of 4 to 6 kt throughout, consistent with the 07:00 UTC routine observation.

No tailwind component is recorded at any point.

The reject was initiated 3 seconds after V1 and at 118 kt, six knots above it.

Power lever movement to flight idle took 0.6 seconds and maximum braking was established 2 seconds after the decision. That response time is at the fast

end of the operator's own recorded fleet distribution for rejected take-offs (fleet mean 3.2 seconds from decision to maximum braking).

Reverse was selected on No. 1 only. No. 2 propeller RPM decays from 07:30:33, consistent with the blade damage found on inspection, and the No. 2 reverse interlock did not make. The scheduled accelerate-stop figures assume symmetric reverse; the achieved mean deceleration of 2.6 m/s² is approximately 15% below the value used in the certified accelerate-stop data for this weight and runway condition.

Distance reconstruction from recorded groundspeed puts the aircraft 1,152 m along runway 09 at the moment of the reject, leaving 695 m of sealed surface. The stopping distance actually achieved from 118 kt was 757 m.

Source: Flight Data Section report FDA-2026-0417-R1, issued 24/06/2026

- No. 2 propeller: three blades damaged beyond repairable limits, leading edge deformation and one blade tip separation. Animal tissue recovered from two blade roots. Engine subject to a shock-load inspection; borescope of both engines found no pre-existing damage and no evidence of any in-flight or on-ground power anomaly before the impact.
- Right main landing gear: both tyres deflated through the fuse plugs, consistent with brake energy. Gear doors distorted by impact. Gear structure within limits on detailed inspection.
- Brakes: brake energy absorbed calculated at 84% of the certified maximum for the weight and speed. No brake fire, no hydraulic leak. Anti-skid system tested serviceable after the occurrence.
- Lower right fuselage skin: abrasion and two punctures consistent with propeller debris.
- Runway and RESA survey: two animal carcasses recovered, one at 1,188 m and one at 1,221 m from the runway 09 threshold. Continuous tyre marks from 1,160 m to the runway end. Nose wheel rut in the RESA to 62 m beyond the end, on the extended centreline.

Source: Engineering inspection report, VH-SBQ, 19/06/2026 to 21/06/2026